LNCT UNIVERSITY, BHOPAL

CSE-602
B.TECH (CSE) VI SEMESTER
EXAMINATION
NATURAL LANGUAGE PROCESSING& Gen AI

Maximum Marks: 70 **Time Allowed: 3 Hours**

Note:- Attempt all questions. Internal choice is given.

SECTION – A

I. Short Answer Type Questions (Attempt Any Five) [5×4=20]

i. Define Natural Language Processing (NLP) and list two real-world applications.
ii. What is the role of Named Entity Recognition (NER) in NLP tasks?
iii. Explain the concept of word embeddings with reference to Word2Vec and GloVe.
iv. Differentiate between BERT and GPT in terms of language modeling approaches.
v. What are the key steps in the NLP project pipeline?
vi. Define Self-Attention and explain its significance in transformer models.
vii. What are deepfakes? Mention any two safety measures to detect or prevent them.

SECTION – B

II. Long Answer Type Questions (Attempt Any Four) [4×10=40]

i. Discuss the various **linguistic foundations** of NLP, such as syntax, semantics, ambiguity, and code-switching, with suitable examples.
ii. Explain **data preprocessing techniques** in NLP. How do tokenization, stopword removal, and normalization improve model performance?
iii. Describe the core components of the **Transformer architecture**. How do encoder-decoder mechanisms function?
iv. Explain **sampling strategies** used in text generation (Greedy, Beam, Top-k, Top-p). Which strategies improve diversity and fluency?
v. Analyze the challenges and risks associated with **Cross-Modal Generative AI**, including synthetic media and misuse. Mention policy-level safety measures.
vi. Compare **Contextual Embeddings** (e.g., BERT, ELMo) with traditional word embeddings (e.g., Word2Vec). How does context influence meaning?